

Algebra 1 — Tutorial sheet 5

Transcribed from the Moodle assignment page (no PDF was posted). Opened 25 Sep 2026 Due Fri 2 Oct 2026. No separate HW subset named.

0. If you have not done the HW 4 computations, finish them and submit, especially 11(a) and (c).
1. Look at the example in the Sep 17-3 board image: the map $f(x) = Ax$ has a simple diagonal matrix with respect to a “magic” basis. Finding such bases is a goal for the next few lectures.
2. An operator T on V is a *projection* if $T^2 = T$. Show that if T is a projection then $V = \text{im } T \oplus \ker T$ (internal direct sum). Is this true if V is infinite-dimensional? The converse is false (pointed out by Pradyun): find a counterexample. What extra condition gives an “if and only if”? Why the name “projection”?
3. Artin, Chapter 3, problems 5.1, 5.2, 5.3 (direct sums).
- 4 (SMMC 2023). Let $n \geq 1$ and let A, B, C be n -dimensional subspaces of $\mathbb{R}^{2n}$ with $A \cap B = B \cap C = C \cap A = \{0\}$. Prove there are bases $\{a_i\}$ of A , $\{b_i\}$ of B , $\{c_i\}$ of C such that for each i the vectors a_i, b_i, c_i are linearly dependent.
5. Verify that the examples in the Sep 17-5 board image are fields. Review that every concept so far goes through over a general field.